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Inferring Planet Compositions Using Statistical Methods

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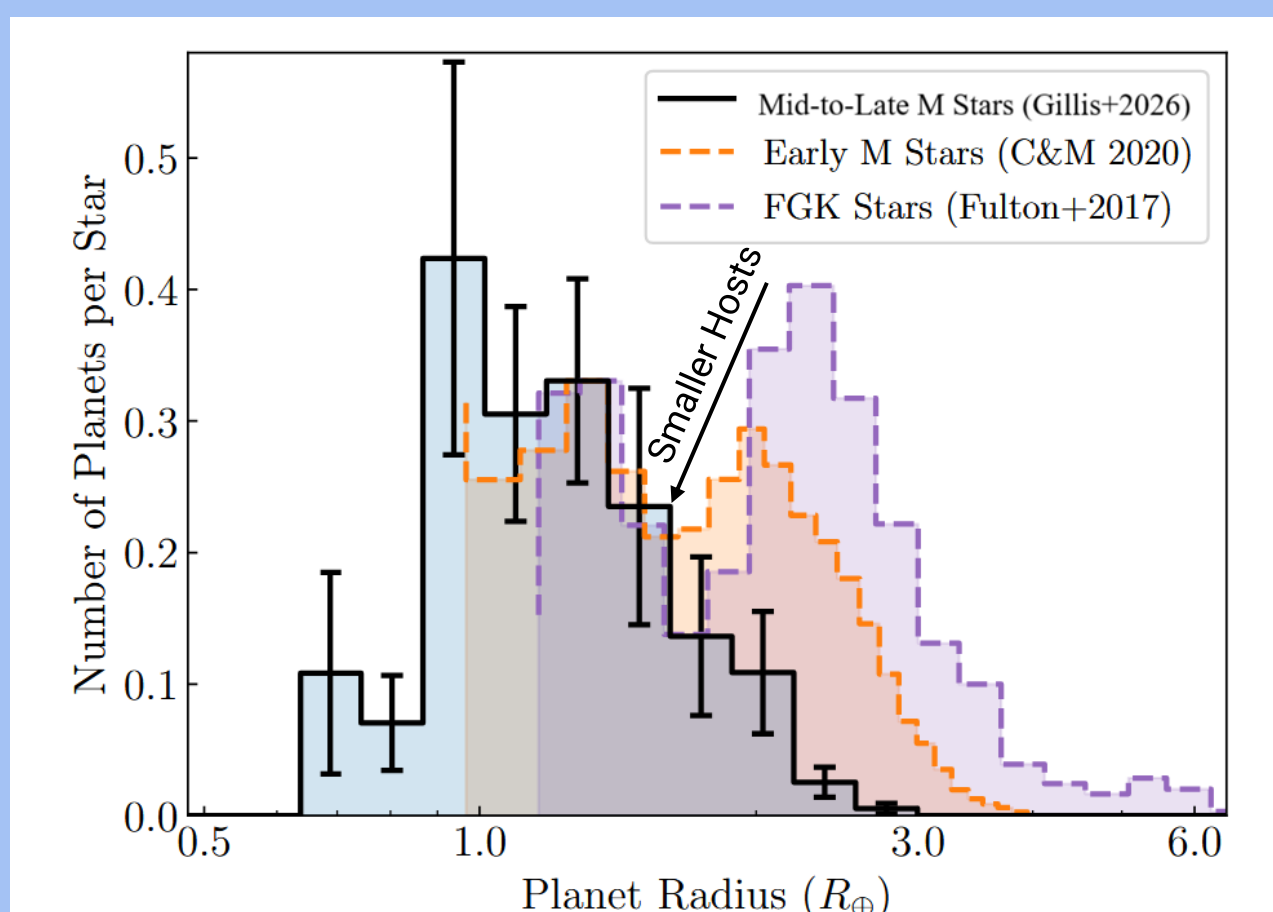
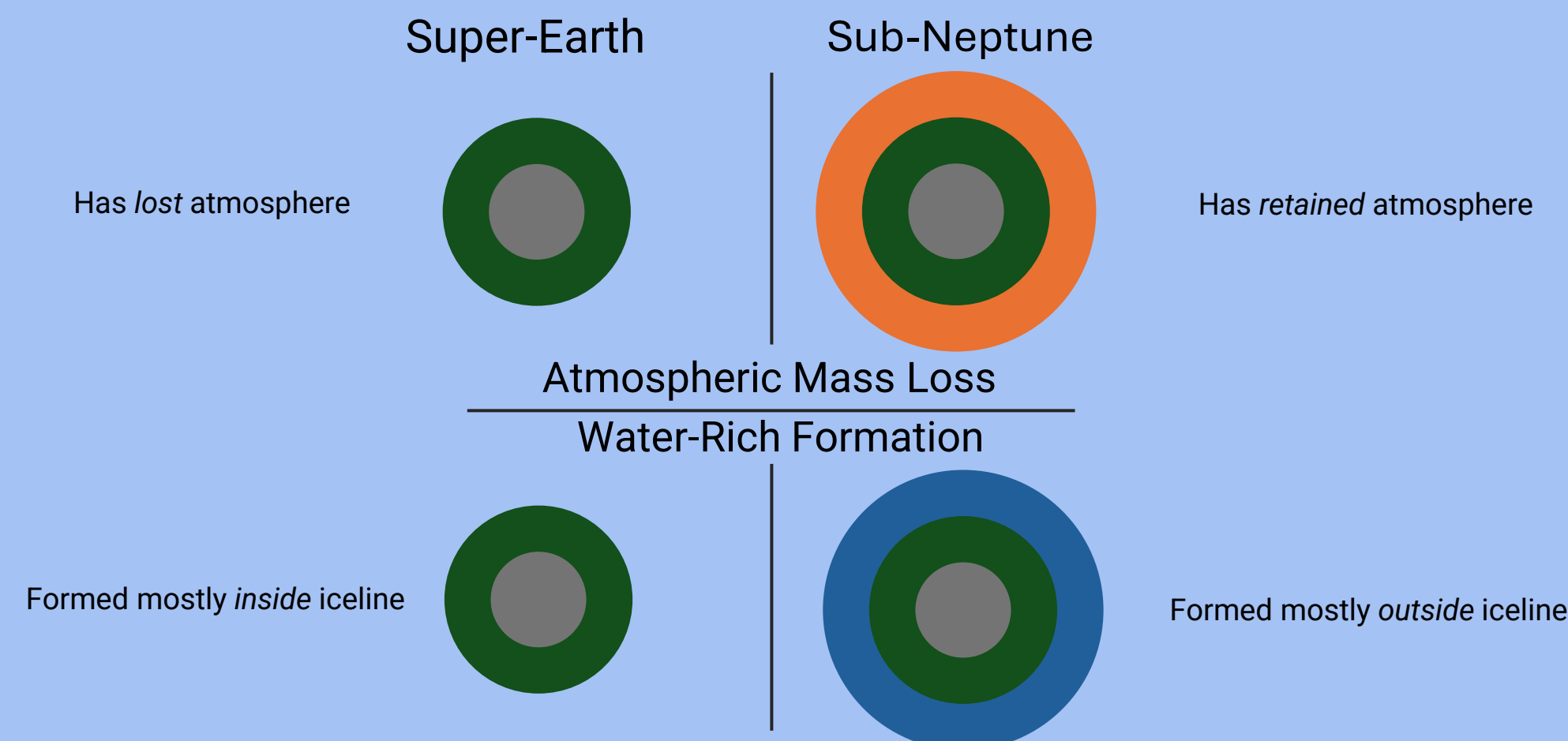


Background

- Observations give planetary masses (M), radii (R), and equilibrium temperatures (T)
- But to compare to theory, need *compositions*, which must be *inferred* from observed parameters using an **Interior Structure Model** that gives $R(M, \text{composition})^1$
- Convert $R(M, T, \text{Composition}) \rightarrow \text{Composition}(M, R, T)$
- Highly degenerate—over a dozen free composition parameters constrained by just 3

The Need For Compositions

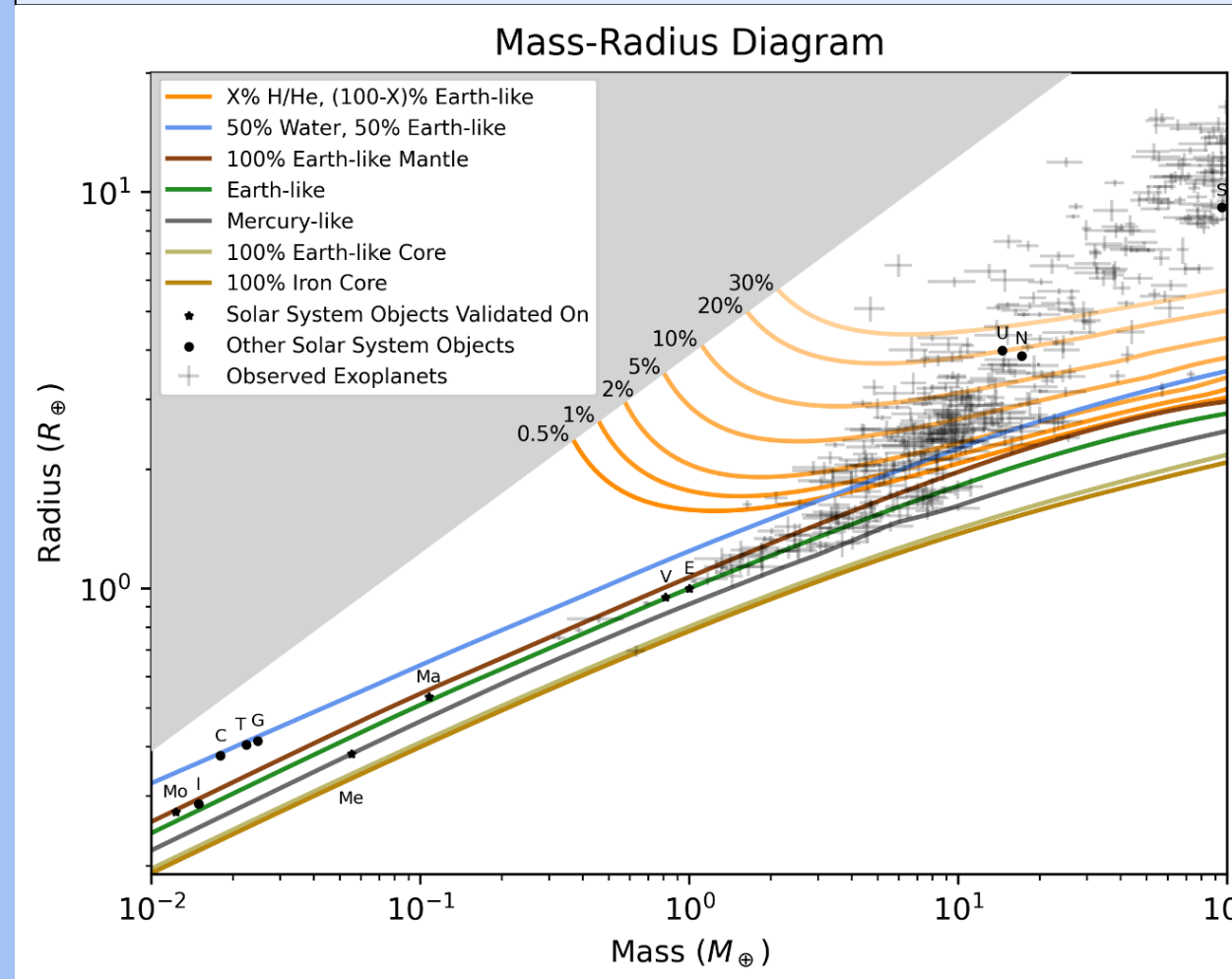
- Planet radii are bimodal around FGK stars, with smaller *super-Earths* and larger *sub-Neptunes*²⁻⁴
- Two possible differentiating mechanisms: atmospheric mass loss⁵ (sub-Neptunes are H/He-enhanced) or water-rich formation⁶ (sub-Neptunes are water-enriched)
- Sub-Neptunes less prominent around lower-mass stars²⁻⁴ -> Finding their composition reveals planet formation around most common type of star
- Sub-Neptunes around M Dwarfs can be closer in -> would have lost atmosphere -> could be water-rich³
- If water-rich -> widespread planet migration⁶



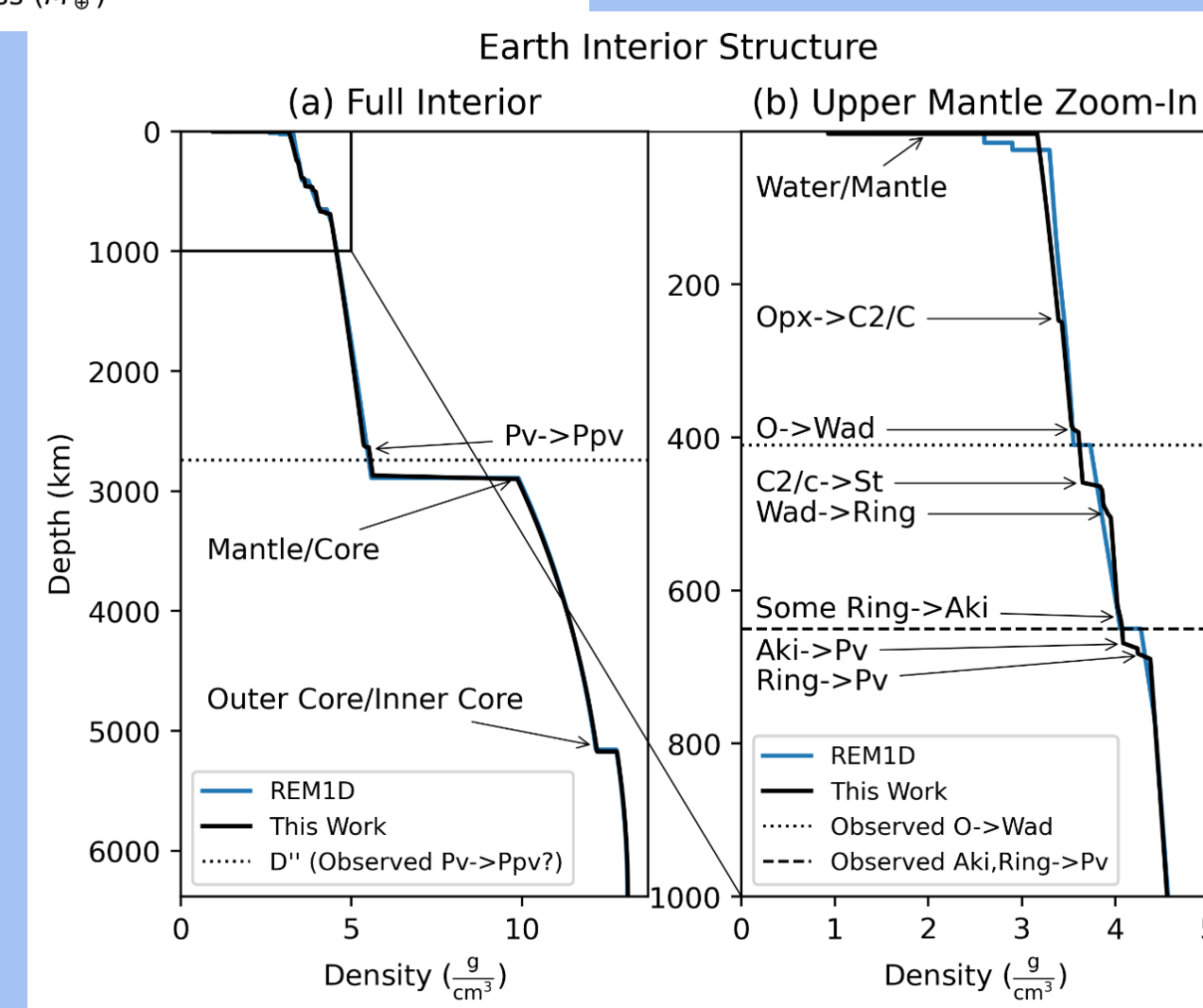
Reproduced from Gillis, Cloutier, & Pass (2026) Fig. 11

A New Forward Model

- **We have created the most comprehensive interior structure model in the literature**¹
- Previous models⁷⁻¹⁸ have systematic biases in model radii -> **literature compositions biased**
 - Bias comparable to observational errors¹⁹, PLATO²⁰ will bring average observational errors further down
- New model uses most up-to-date Equations of State
 - H/He: Non-ideal mixing, new simulation results²¹⁻²³
 - Water: 12+ phases, high P simulations²⁴
 - Mantle: Mixing, melting, high P phases²⁵⁻³²
 - Core: Light element, better high P extrapolation³³⁻³⁷
- Additional corrections: Radiative transfer³⁸⁻³⁹, internal luminosity⁴⁰, transit geometry⁴¹, rotation⁴²⁻⁴³
- Extremely high accuracy solar system replication^{1,44-45}
- Net effect: Cold planets denser, hot planets less dense
- Mass-Radius curves generated; available online



Reproduced from **Skinner**, Pudritz, & Cloutier (2026) Fig. 8

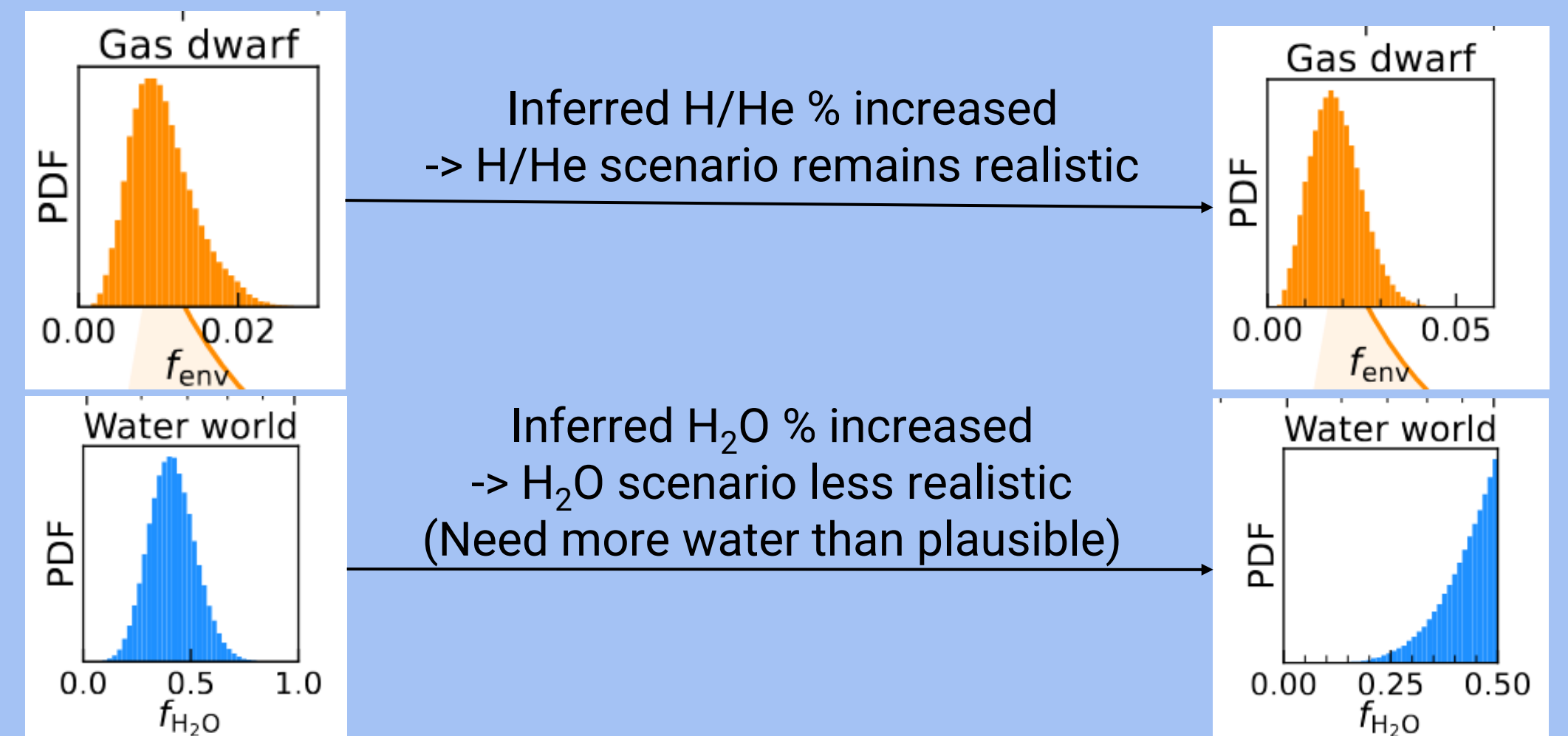


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Applying the New Model

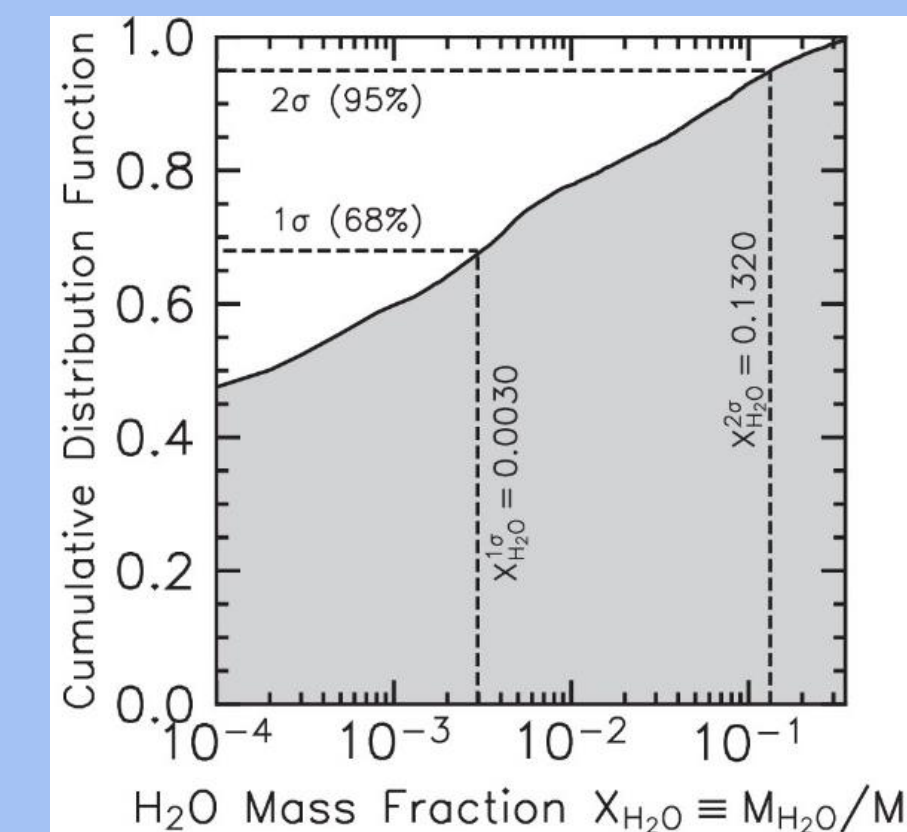
- Incorporated into SMINT framework⁴³⁻⁴⁴
 - Linearly interpolates R from M-composition grid
 - Runs MCMC to retrieve planets' composition
- First application to TOI-210 b makes water-rich composition *less likely*⁴⁴
- Application to more individual targets forthcoming – **email w/ requests to generate grids for specific objects**



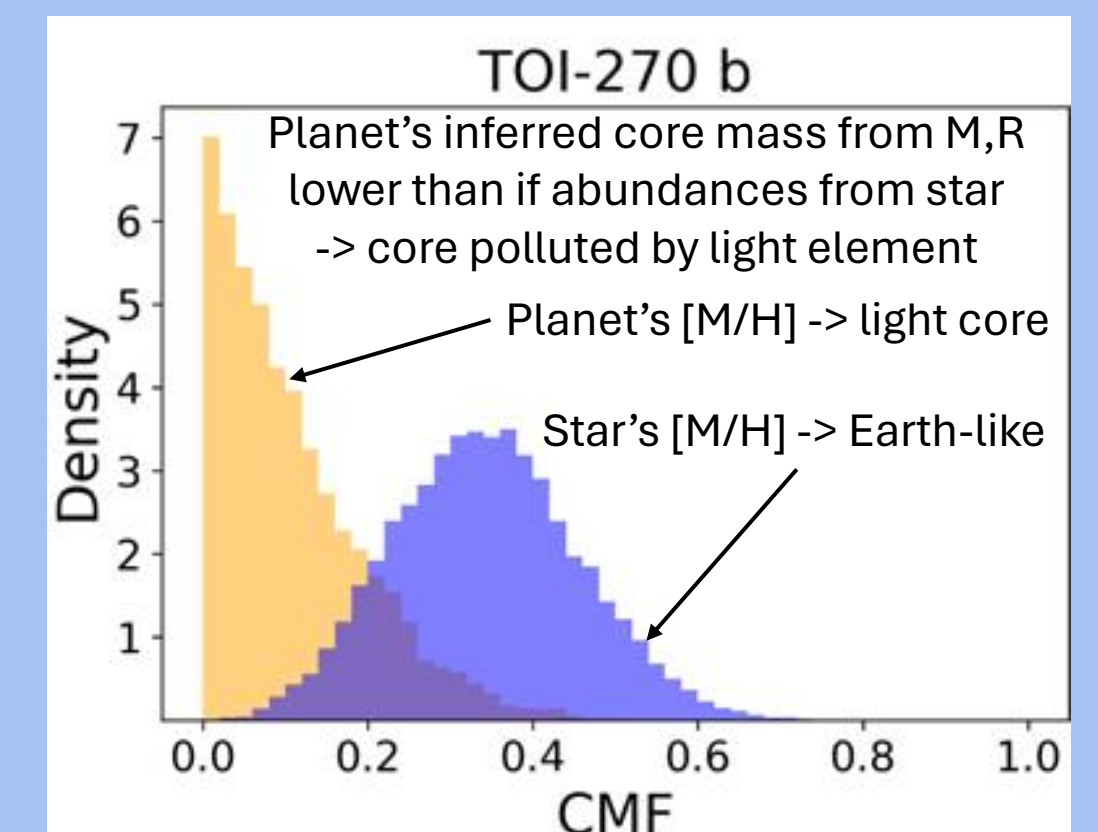
Reproduced from Cadieux, **Skinner**, et al. (In Prep.) Fig. 7 (left previous version, right current)

Recent Literature Advances

- FGK Super-Earths -> Earth-like⁴⁵⁻⁴⁷
- M Dwarf Super-Earths -> Water-rich interiors?⁴⁸



Reproduced from Rogers et al. (2025) Figure 5



Reproduced from Weissman et al. (2026) Figure 7

Future Work

- Integrate our state-of-the-art forward model into a robust statistical model, run on a large population
 - Investigate FGK v. M Dwarf super-Earth dichotomy
 - Conduct first comprehensive sub-Neptune inference
- Possible widely-used methods each w/ downsides⁴⁹⁻⁵¹:
- Interpolation – Inaccurate b/c not directly calculating
- MCMC – computationally intensive (~10 cpu-d/planet)
- Machine learning – Long training makes updates hard
- **We are in the beginning stages of implementing statistical methods – feedback very welcome!**